

UNIT 6

Application Layer

6.1 Functions of Application Layer

- 1. File transfer, access and management (FTAM):** It provides facilities so that a user can access and retrieve files in a remote computer for local usage, as well as manage and store files on a remote computer from the local computer.
- 2. E-mail services:** It provides e-mail services.
- 3. Directory services:** It helps to retrieve global information and services by providing access to distributed database resources.
- 4. Virtual terminal:** It provides a virtual network terminal through which a user can log on to the remote host on the network.

6.2 Application Layer Protocols

1. Domain Name System (DNS):

1. The Domain Name System (DNS) is the phonebook of the Internet. Humans access information online through domain names, like smc.edu.np or espn.com. Web browsers interact through Internet Protocol (IP) addresses. **DNS translates domain names to IP addresses so browsers can load Internet resources.**

2. Each device connected to the Internet has a unique IP address, which other machines use to find the device. DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in IPv4), or more complex newer alphanumeric IP addresses such as 2400:cb00:2048:1::c629:d7a2 (in IPv6).

3. DNS protocol uses the port number **53**.

Working of DNS:

1. The process of DNS resolution involves converting a hostname (such as `www.example.com`) into a computer-friendly IP address (such as `192.168.1.1`).

An IP address is given to each device on the Internet, and that address is necessary to find the appropriate Internet device, just as a street address is used to find a particular home.

When a user wants to load a webpage, a translation must occur between what a user types into their web browser (`example.com`) and the machine-friendly address necessary to locate the `example.com` webpage.

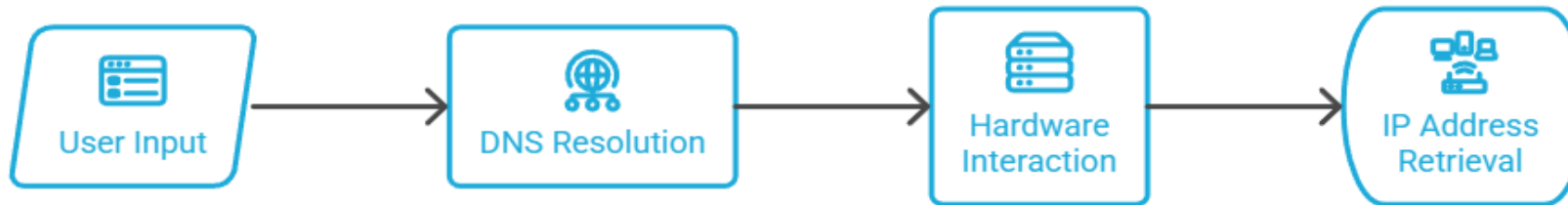
Working of DNS:

2. To understand the process behind the DNS resolution, it's important to learn about the different hardware components a DNS query must pass through.

For the web browser, the DNS lookup occurs “ behind the scenes” and requires no interaction from the user's computer apart from the initial request.

Working of DNS:

DNS Resolution Process Overview



2. Dynamic Host Configuration Protocol (DHCP):

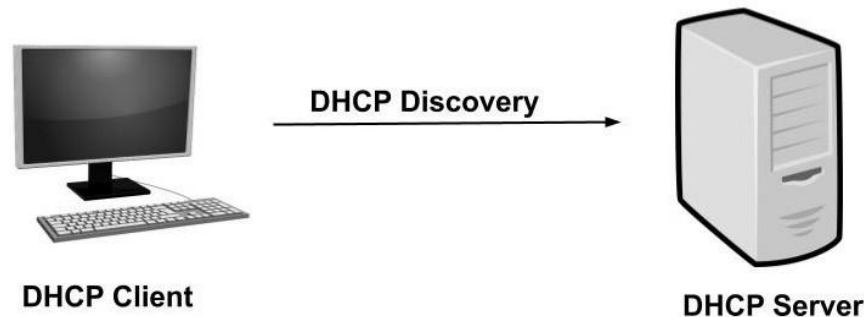
1. DHCP (Dynamic Host Configuration Protocol) is a network management protocol used to dynamically assign an Internet Protocol (IP) address to any device, or node, on a network so they can communicate using IP.
2. DHCP automates and centrally manages these configurations, rather than requiring network administrators to manually assign IP addresses to all network devices.
3. DHCP can be implemented on small local networks, as well as large enterprise networks.
4. DHCP uses port numbers **67 and 68**.

Messages that are used in DHCP operation (DORA):

- 1. Discover:** The DHCP client broadcasts this message to find a DHCP server.
- 2. Offer:** The DHCP server broadcasts this message to lease an IP configuration to the DHCP client.
- 3. Request:** The DHCP client uses this message to notify the DHCP server whether it accepts the proposed IP configuration or not.
- 4. Acknowledgement:** The DHCP server uses this message to confirm the DHCP client that it can use the offered IP configuration.

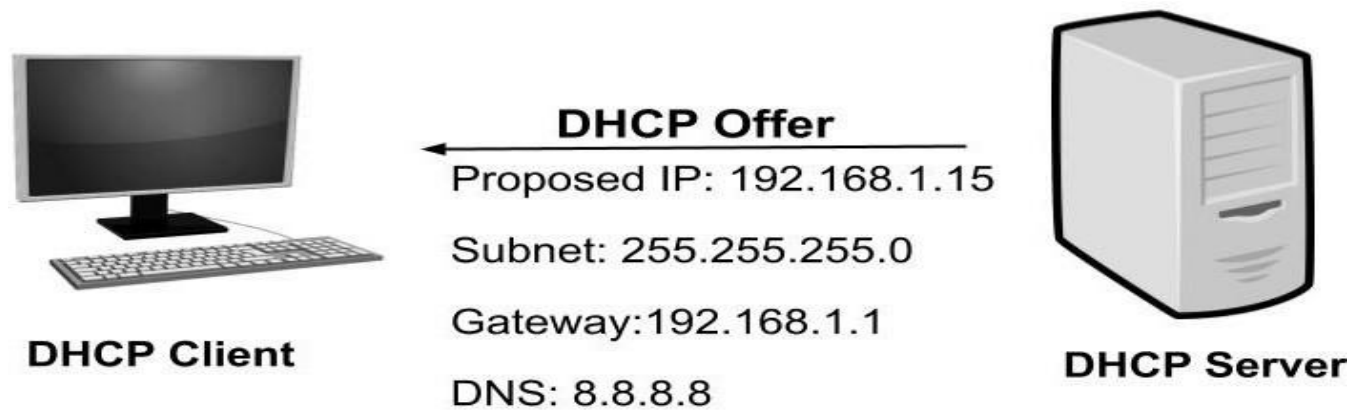
Working of DHCP:

1. DHCP Discovery: The DHCP client broadcast messages to discover the DHCP servers. The client computer sends a packet with the default broadcast destination of 255.255.255.255, which is a special broadcast address, which means “this network”.



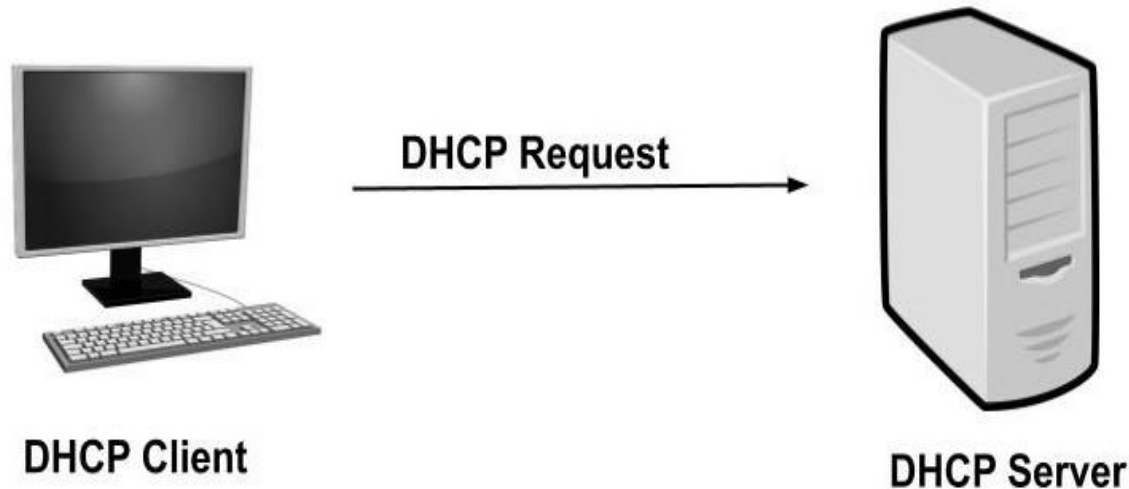
Working of DHCP:

2. DHCP Offer: When the DHCP server receives the DHCP Discover message then it offers an IP address to the client by sending a DHCP offer message to the client. This DHCP offer message contains the proposed IP address for DHCP client, subnet mask, default gateway and DNS address.



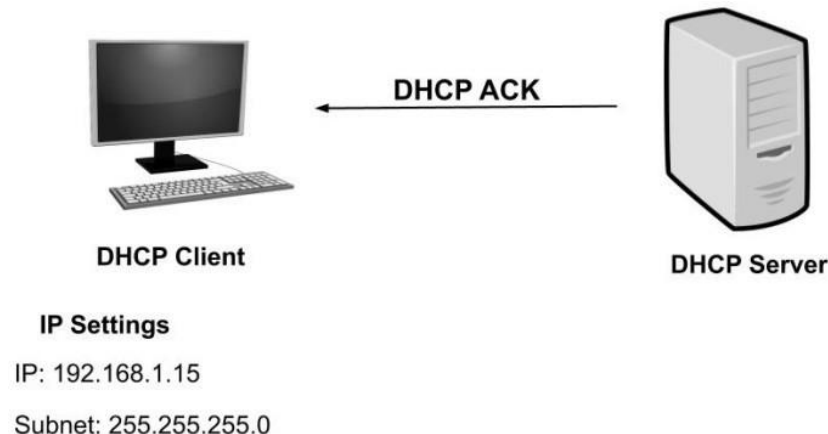
Working of DHCP:

3. DHCP Request: In most cases, the client can receive multiple DHCP offers because in a network, there are many DHCP servers. If the IP addressing of one server fails, then other servers can provide backup. But the client will accept only one DHCP offer. In response to the offer, the client sends a DHCP Request requesting the offered address from one of the DHCP servers.



Working of DHCP:

4. DHCP Acknowledgment: The server then sends an Acknowledgment to the client, confirming the DHCP lease to the client. The server might send any other configuration that the client may have asked. At this step, the IP configuration is completed, and the client can use the new IP settings.



Advantages of DHCP

- It is easy to implement and automatic assignment of an IP address means an accurate IP address.
- The manual configuration of the IP address is not required. Hence, it saves time and workload for the network administrators.
- Duplicate or invalid IP assignments are not there which means there is no IP address conflict.

Disadvantages of DHCP

- As the DHCP servers have no secure mechanism for the authentication of the client, any new client can join the network.
- This poses security risks like unauthorized clients being given IP address and IP address depletion from unauthorized clients.
- The DHCP server can be a single point of failure if the network has only one DHCP server

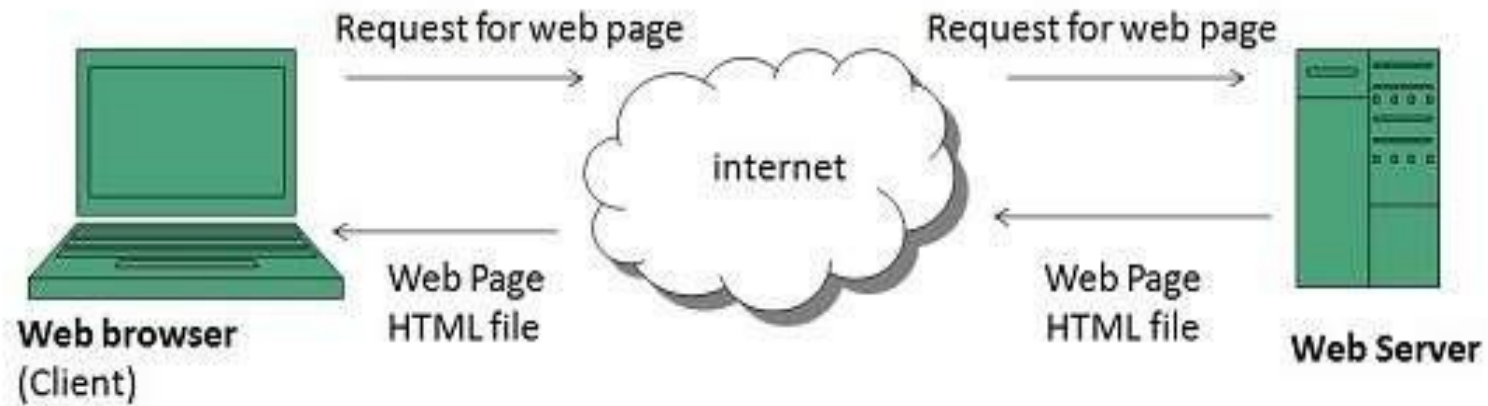
3. World Wide Web (WWW):

- WWW stands for World Wide Web.
- The World Wide Web was created by Timothy Berners Lee in 1989 at CERN (European Council for Nuclear Research) in Geneva.
- A technical definition of the World Wide Web is : all the resources and users on the Internet that are using the Hypertext Transfer Protocol (HTTP).
- In simple terms, the World Wide Web is a way of exchanging information between computers on the Internet, tying them together into a vast collection of interactive multimedia resources.

WWW Operation:

- WWW works on a client-server approach. Following steps explain how the web works:
- User enters the URL (say, <http://www.facebook.com>) of the web page in the address bar of The web browser.
- Then the browser requests the Domain Name Server for the IP address corresponding to www.facebook.com.
- After receiving an IP address, the browser sends the request for web page to the web server using the HTTP Protocol, which specifies the way the browser and web server communicate.
- Then the web server receives a request using the HTTP protocol and checks its search for the requested
Web page. If found, it returns it to the web browser and closes the HTTP connection.
- Now the web browser receives the web page, it interprets it, and displays the contents of the web page In the web browser's window.

WWW Operation:



4. HyperText Transfer Protocol (HTTP) and HyperText Transfer

Protocol Secure (HTTPS):

1. HyperText Transfer Protocol (HTTP) :

HyperText Transfer Protocol (HTTP) is a protocol used to transfer hypertext over the Web. Due to its simplicity, HTTP has been the most widely used protocol for data transfer over the Web, but the data (i.e., hypertext) exchanged using HTTP isn't as secure as we would like it to be. In fact, hypertext exchanged using HTTP goes as plain text, i.e., anyone between the browser and server can read it relatively easily if one intercepts this exchange of data.

4. HyperText Transfer Protocol (HTTP) and HyperText Transfer

- **2. Hypertext Transfer Protocol Secure (HTTPS) :**
- Hypertext Transfer Protocol Secure (HTTPS) is an extension of the Hypertext
- Transfer Protocol (HTTP). It is used for secure communication. In HTTPS, the communication protocol is encrypted using Transport Layer Security.



HTTP vs HTTPS:

S.No	HTTP	HTTPS
1.	HTTP stands for HyperText Transfer Protocol.	HTTPS for HyperText Transfer Protocol Secure.
2.	In HTTP, URL begins with “http://”.	In HTTPS, URL starts with “https://”.
3.	HTTP uses port number 80 for communication.	HTTPS uses 443 port number for communication.
4.	HTTP is considered to be unsecure.	HTTPS is considered as secure.
5.	In HTTP, Encryption is absent.	Encryption is present in HTTPS.
6.	HTTP does not require any certificates.	HTTPS needs SSL Certificates.

5. Telnet:

TELNET stands for Terminal Network. It is a type of protocol that enables one computer to connect to a local computer.

It is used as a standard TCP/IP protocol for virtual terminal service, which is given by ISO. Computer which starts a connection is known as the local computer.

A computer that is being connected to, i.e., which accepts the connection known as a remote computer.

During telnet operation, whatever is being performed on the remote computer will be displayed by the local computer. Telnet operates on client/server principle.

Local computer uses a telnet client program, and the remote computers use a telnet server program. It typically runs on port 23.

6. File Transfer Protocol (FTP):

- FTP represents File transfer protocol and it is a standard internet protocol supported by TCP/IP used for transmitting the files from one host to another.
- FTP needs TCP as a transport protocol to help the reliable end-to-end connections and executes two types of connections(control connection and data connection) in managing data transfers.
- For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc., FTP makes use of control connection. The control connection is initiated on port number 21.
- For sending the actual file, FTP makes use of data connection. A data connection is initiated on port number 20.

Advantages of FTP:

The advantages of FTP are as follows –

- Speed – The FTP is one of the quickest ways to transfer documents from one device to another.
- Security – It can create the FTP server. We need to log in with the username and password.
- Back & forth movement – FTP enables us to send the files back and forth.

Disadvantages of FTP:

- The standard requirement of the market is that all FTP transmissions should be encrypted. However, not all FTP providers are equal and not all providers support encryption.
- Passwords and file text are sent in clear text that enables unwanted eavesdropping. Therefore, it is quite possible that attackers can carry out a brute force attack by trying to guess the FTP password.

7. Simple Mail Transfer Protocol (SMTP):

- Most internet systems use SMTP as a method to transfer mail from one user to another. SMTP is a push protocol and is used to send the mail whereas POP (post office protocol) or IMAP (internet message access protocol) are used to retrieve (pull) those mails at the receiver's side.
- SMTP is an application layer protocol. The client who wants to send the mail opens a TCP connection to the SMTP server and then sends the mail across the connection. The SMTP server is always on listening mode. As soon as it listens for a TCP connection from any client, the SMTP process initiates a connection through port 25. After successfully establishing a TCP connection the client process sends the mail instantly.

The SMTP model is of two type :

- End-to-end method
- Store-and-forward method
- The end to end model is used to communicate between different organizations

whereas the store and forward method is used within an organization.

8. Post Office Protocol Version 3 (POP3):

- It is a standard mail protocol used to receive emails from a remote server to a local email client.
- It allows us to download email messages on our local computer and read them even when we are offline and removed from the email server.
- By default, the POP3 protocol works on two ports:
- Port 110: non-encrypted port
- Port 995: encrypted port
- Mail access starts with the client when the user needs to download the received email from the mailbox on the mail server.
- The client sends the user's name and password to access the mailbox and then can list and retrieve the mail messages.

9. Internet Message Access Protocol, Version 4 (IMAP4):

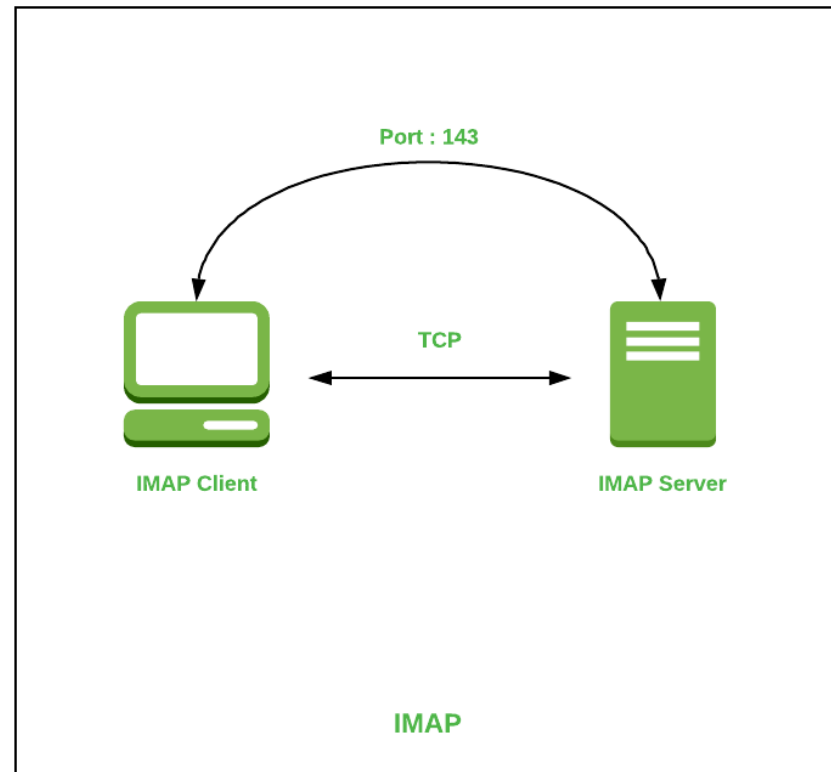
- IMAP is an application-layer protocol that allows us to access our email messages wherever we are; much of the time, it is accessed via the Internet.
- The current version of IMAP is IMAP4.
- It works on port 143 by default, which is non-encrypted, and for secure encrypted communication port, 993 is used.

Advantages:

- Users have remote access to all the contents.
- It offers synchronization across all the maintained sessions by the user.

Disadvantages:

- IMAP is complex to maintain.
- It is slower to load messages.



6.3 Concept of Network Traffic Analyzer

- Network Traffic Analyzer is an application to record and analyze the traffic on the network.
- It can break down the traffic by application, user, or IP address.
- The tool visualizes the data flows through diagrams or tables.
- Network Traffic Analysis tools can automatically collect data, display it in a visual format, send alerts, provide reports and correlate data from across the network.
- The process provides the network security by detecting anomalies in the network behavior.

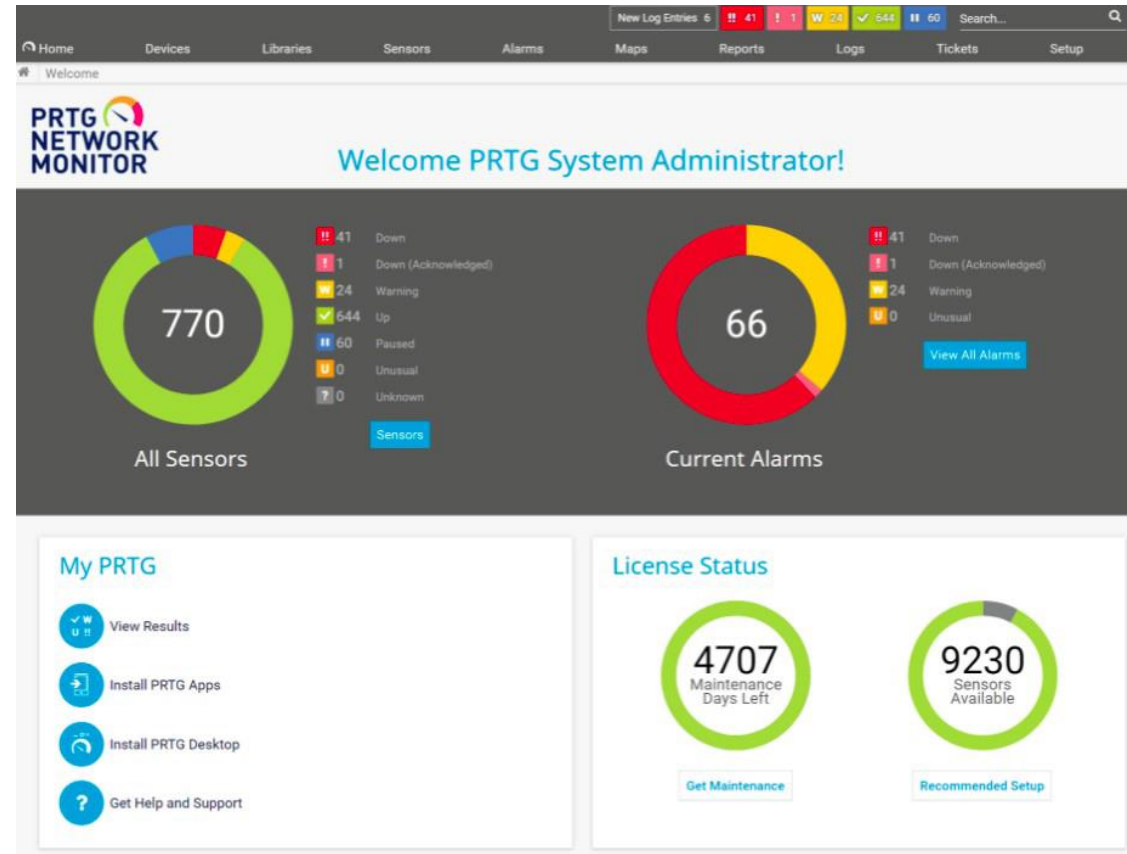
1. Multi Router Traffic Grapher (MRTG):

- **Multi Router Traffic Grapher (MRTG)** is a network monitoring tool used to monitor and graph the traffic load on network links and devices such as routers, switches, and servers.
- Monitors network traffic in real time.
- Generates graphical reports of bandwidth usage.
- Uses SNMP to collect network statistics.
- Displays daily, weekly, monthly, and yearly traffic trends.
- Helps identify network congestion and performance issues.

2. Paessler Router Traffic Grapher (PRTG):

- **PRTG (Paessler Router Traffic Grapher)** is a comprehensive **network monitoring tool** developed by [Paessler AG](#).
- Monitors network traffic and bandwidth usage.
- Provides real-time alerts and notifications.
- Displays data through customizable dashboards and graphs.
- Monitors servers, routers, switches, applications, and cloud services.

2. Paessler Router Traffic Grapher (PRTG):



3. Simple Network Management Protocol (SNMP):

- If an organization has 1000 of devices then to check all devices, one by one every day, are working properly or not is a hectic task.

To ease these up, Simple Network Management Protocol (SNMP) is used.

- SNMP is an application layer protocol which uses UDP port number 161/162.

SNMP is used to monitor the network, detect network faults and sometimes even used to configure remote devices.

SNMP components –

There are 3 components of SNMP:

(a) SNMP Manager –

It is a centralized system used to monitor network. It is also known as Network Management

Station (NMS)

(b) SNMP agent –

It is a software management module installed on a managed device. Managed devices can be

network devices like PC, router, switches, servers etc.

(c) Management Information Base –

MIB consists of information of resources that are to be managed. These information is organized

hierarchically. It consists of objects instances which are essentially variables

4. Cisco Packet Tracer (CPT):

Cisco Packet Tracer (CPT) is a **network simulation and visualization tool** developed by [Cisco Networking Academy](https://www.cisco.com/go/netacad).

It allows students and network professionals to design, configure, simulate, and troubleshoot computer networks in a virtual environment without requiring physical networking devices.

Features of Cisco Packet Tracer

- Simulates routers, switches, PCs, servers, and wireless devices.
- Supports network design and configuration.
- Provides real-time and simulation modes.
- Allows testing of network protocols and services.
- Supports IoT (Internet of Things) and basic cybersecurity simulations.

Key Features:

- Unlimited devices
- E-learning
- Customize single/multi user activities
- Interactive Environment
- Visualizing Networks
- Real-time mode and Simulation mode
- Self-paced
- Supports majority of networking protocols
- International language support
- Cross platform compatibility

5. WireShark:

- **Wireshark** is a **network protocol analyzer** used to capture, monitor, and analyze network traffic in real time. It helps network administrators, security analysts, and students understand how data travels across a network.

Functions of Wireshark

- Captures network packets.
- Analyzes network protocols such as TCP, UDP, HTTP, DNS, and ICMP.
- Troubleshoots network problems.
- Detects security threats and suspicious traffic.
- Monitors network performance.